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Optimizing Highly Deviated Wells in Egypt: Cost-Effective Transition from High-Power ESP to Reciprocating Rod Pumping with Continuous Rod

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Abstract

In Egypt's oil and gas industry, rod-driven artificial lift systems (ALS) are traditionally considered suitable only for vertical wells. Electrical Submersible Pumps (ESP) has been the preferred choice for highly deviated wells due to the high costs associated with tubing leaks and sucker rod failures. However, ESPs are expensive and consume significant power. Over time, the operational expenses of ESPs can become prohibitive, making the well economically unviable due to high power consumption and costly workover interventions. This study proposes an optimal ALS solution to reduce power consumption, workover intervention costs, and lifting expenses in Egypt's highly deviated wells. The BED 3-26 well is a highly deviated and deep well with a dogleg severity of 8.44°/100ft and a side load of 900 lbf. Producing approximately 250 barrels of fluid per day (BFPD) from perforation depths around 10,000 feet, this well has been equipped with an ESP since 2020 due to its high DLS value and depth. A comprehensive investigation, including production history, workover history, deviation survey, and intervention reports, revealed that ESP cable failure is the primary cause of failure, accounting for over 95% of incidents, with the remaining 5% due to pump broken shafts. The maximum run life was 90 days. The high operating costs of ESPs, due to frequent workover and high-power requirements, prompted the operator to seek a more cost-effective solution. This research demonstrates that combining Reciprocating Rod Pumping (RRP) with continuous rod is an effective approach for overcoming the limitations of rod-driven ALS in aggressive wellbore geometries in Egypt. The results show that transitioning from an expensive ESP system to a more cost-effective RRP with continuous rod solution can expedite the return on investment. Compared to previous lift systems, this solution reduced capital expenses by 39%, flow losses by 41%, power usage by 70%, and increased the time between failures by 411%. The well has been operating continuously, producing 250 BFPD for 460 days without any costly interventions, compared to 90 days with the ESP system.

Keywords: ESP, Driven Rod, RPP, Continuous Rod, Side Load, Tubing Wear, Power Consumption, Workover Interventions, Highly Deviated Wells

Introduction

The Badr-3 (BED-3) field is a significant asset of Badr El-Din Petroleum Company (BAPETCO), located in Egypt's Western Desert within the Abu Gharadig Basin. This field is part of the Badr El-Din (BED) concession, a joint venture between Shell Egypt and the Egyptian General Petroleum Corporation (EGPC). BED-3 is primarily an oil-producing field, with substantial reserves in the Abu Roash G formation. The field's development has been marked by significant drilling activities, including the successful drilling of the Bed3C6-1 well, which was tested at a rate of 3,850 barrels per day. This well targeted the hydrocarbon potential of the Abu Roash G formation, revealing a thickness of approximately 46 ft with 24% porosity in the upper sandstone reservoir, and 40 ft with 20% porosity in the lower sandstone reservoir. The BED-3 field is situated between latitudes 29°45' and 30°05'N and longitudes 27°30' and 28°10'E, occupying the northwestern part of the Abu Gharadig Basin. The field is characterized by a WNW–ESE trending horst structure, dipping to the east. The main pay zone in the field is the Albian–Cenomanian Kharita Formation, which is a high-energy shallow marine shelf environment. The Badr-3 field is integral to BAPETCO's strategy of leveraging advanced technologies and drilling techniques to maximize hydrocarbon recovery and ensure a sustainable production profile.

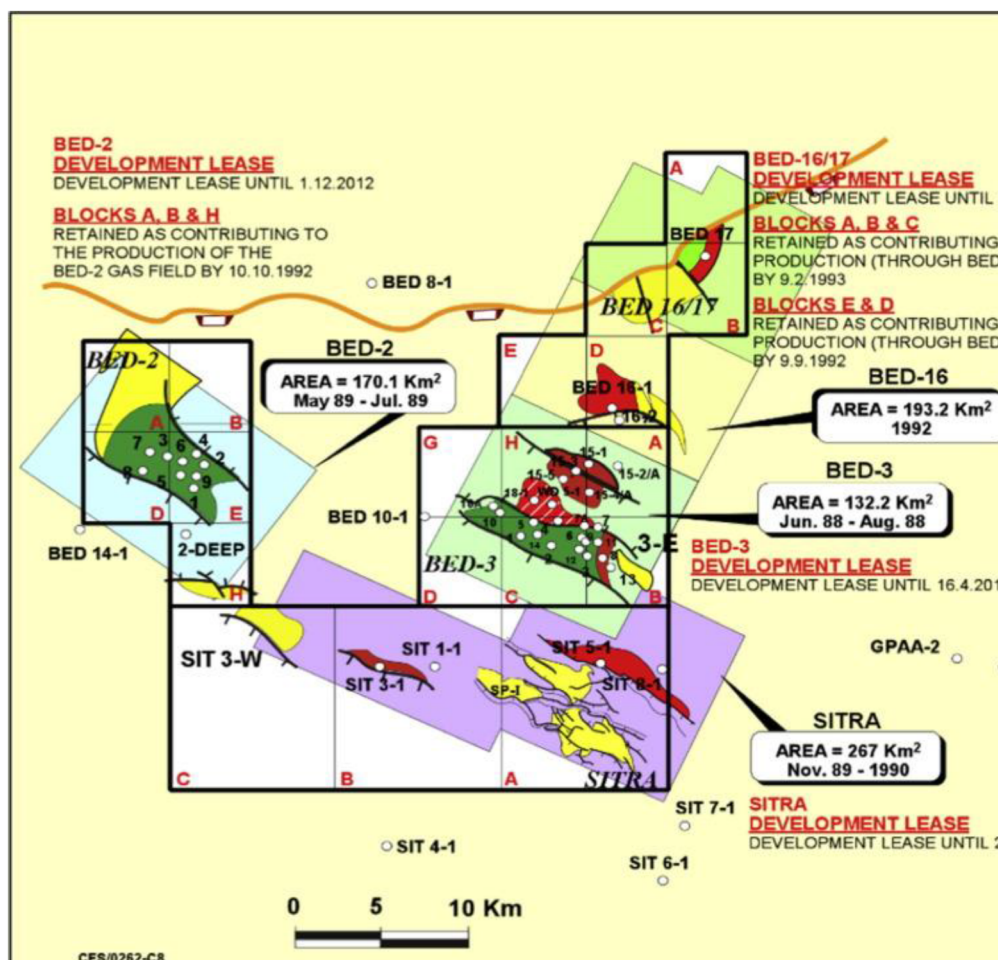


Figure (1)—Location map of Badr fields

In Egypt's oil and gas sector, rod-driven artificial lift systems are typically recommended for vertical or deviated wells with low DLS values ($\leq 3^\circ/100$ ft). For highly deviated wells, operators often prefer electrical submersible pumps (ESPs) to avoid interventions related to tubing or sucker rod failures. However, ESPs are not always the optimal solution for all applications or well depths due to their high capital

expenditure (CAPEX) and significant power consumption. Rod-driven artificial lift systems can be a more economical alternative, particularly for low-rate wells. At flow rates below 1,000 STB/D, ESPs exhibit low efficiencies (less than 50% at their highest efficiency point), leading to increased power consumption and heat generation, which in turn results in higher costs and shorter operating times. Additionally, changing well conditions can cause the ESP operating point to shift away from the recommended operating range, resulting in lower efficiency and reduced run life. The operating expenses of ESPs in depleted reservoirs are high because downhole equipment failures typically necessitate workover. Given that the majority of ESP capital costs are associated with downhole equipment, several Egyptian companies are seeking more economical solutions for these challenging applications. This study presents a successful solution for this problem in one of Egypt's oil fields, specifically well BED 3-26. Well BED 3-26 is located in the BED-3 field in Egypt's Western Desert, approximately 300 kilometers west of Cairo. The middle of the perforations is approximately 10,000 feet. Most well interventions have been caused by ESP cable failures. Various solutions have been developed to extend the time between failures, but the maximum run life achieved is 90 days.

Literature Review

Because rod-driven ALS is commonly employed in oil production, it is essential to analyze the system as a whole to minimize the system size and energy consumption for lifting oil, improving pumping efficiency, and maximizing the production rate. Increasing pump life and reducing downtime or operating expenses also depend on reducing the frequency of system failures and the number of maintenances. The sucker rod is an essential component in determining the pump maintenance frequency, power consumption, and optimal operating parameters. Several researchers have made efforts to identify the causes of sucker rod failures. Mueller and O'Neil (1969) [1] discussed initiatives to prevent sucker rod failures in the South Glenrock Field and Muddy Reservoir. Several engineering and operational solutions have been employed, including addressing coupling materials and makeup, designing a high-volume lift program, and detecting fatigue cracks via magnetic particle inspection. Maston (1971) [2] explained how the oil and gas industry can replace sucker rods with more efficient alternatives. He proposed a graphical approach and mathematical formula for estimating the optimal point at which the cost of replacing a worn-out rod string exceeds its maintenance costs. He also investigated how these strategies could be applied to decisions on partial rod string replacement. Ditmore (1973) [3] discussed the field operations expertise of the Getty Oil Company with new sucker rods of ultrahigh strength. The new rods have an enhanced capacity for rod pumping in the field, allowing for the use of larger pumps. Mohamed (1981) [4] explained the causes of sucker rod failures in the oil industry and how to avoid them. He emphasized that incorrect handling, improper application, or service-related issues account for the majority of sucker rod failures rather than defective steel. He also clarified the causes of fatigue failure, such as stress magnitude, corrosive environment, surface condition, and stress range, as well as the fact that fatigue is the primary cause of rod failures. Allen and Svinos (1984) [5] developed a program to lower rod and tube failure rates in Crane County fields in Texas. The program included procedures for replacing broken rods and tubing, utilizing inspected and categorized rods and tubing, and employing high-stress rods and fiberglass sucker rod systems. The article also emphasized the significance of suitable rod make-up and investigated the redesign of power tongs for coupling make-up. McCaslin (1987) [6] presented a study on preventing rod-wear tube leaks in sucker rod pumping wells. He observed two primary causes of these leaks: rapid corrosion and significant hole deviation. He emphasized that the key to preventing these leaks is to properly identify whether they are caused by mechanical or corrosion wear and then perform the necessary repairs. He also discussed the use of soft rod guides, rod rotators, and oil film lubrication to reduce mechanical wear. Murtha (1987) [7] investigated the environmental, wear, and impact properties of a sucker rod guide made of the composite material PPS (polyphenylene sulfide). The paper provides field and laboratory test results demonstrating the

high impact strength and useful wear properties of PPS composite materials. Ryan (1988) [8] discussed the advantages of employing spray metal plastic-coated (SMPC) sucker rods at West Texas production facilities for rod body corrosion reduction. He presented information on the operating histories of approximately 650 bare steel and SMPC rod strings installed since 1977. The document showed that SMPC sucker rods were successful at reducing corrosion-related rod failures and increasing sucker rod string life. O. Rivas et al. (1990) [9] suggested the development of a sucker rod centralizer to reduce wear and increase the life of the tubing, sucker rod string, and subsurface pump. The centralizers were tested in seven deviated wells in the Orinoco Belt Field, which resulted in a longer active life of the equipment. Matthews and Dunn (1993) [10] presented research performed in the Albertan Lindberg heavy-oil field to evaluate wear-related problems in directional wells and develop strategies to extend the life of production equipment. The study monitored production characteristics as well as rod and tubing wear in more than 50 directional wells. The study's findings showed that the wellbore configuration and operational parameters had a substantial impact on how rapidly downhole equipment was worn down in directional wells. Heinze and Rahman (1999) [11] examined data from more than 25,000 sucker rod pumping wells in the Permian Basin to identify failure causes and make decision-making recommendations. The study covered methods for organizing and interpreting failure data, as well as comparisons by year, company, location, area, reason for failure, and frequency. The study showed that pump failures accounted for 40% of the failures, followed by rod failures (32%) and tubing failures (28%). Carstensen (2004) [12] identified the root causes of sucker rod connection failures, such as improper makeup and work cycle fatigue. The research highlighted the importance of a well-designed connection system for extending working life while reducing associated costs. Moises, Andrade, and Garcia (2010) [13] discussed the development and implementation of a sucker-rod pumping failure diagnostic system and ontology, emphasizing its ability to extend the MTBF of sucker rod pumping wells and detect pumping system failures. Moustafa and Greer (2011) [14] emphasized the need to properly design and select completion components for beam-pumped wells to extend their life and reduce operating expenses. These authors emphasized the significance of considering the formation depth, total well depth, well variation, fluid properties, necessary production, and sand and gas output when designing completions. Ghareeb, Ellaithy, and Zahran (2012) [15] provided useful insights into the usage of artificial lift systems in Egyptian oil fields, with particular focus on the challenges that were faced and the steps taken to overcome them. This study stressed the need to select the proper lift system based on well-defined criteria, as well as the need for continuous improvement and collaboration with suppliers to increase system performance and productivity. Ali, Mahmoud and Mahmoud (2015) [16] investigated the optimization of artificial lifting systems at the Sahara Oil and Gas's (SOG) West Qarun and North Bahariya oil fields in Egypt's Western Desert. They emphasized the significance of continual optimization and periodic evaluation of sucker rod pumping systems to achieve maximum pumping rates, minimize operational costs, and reduce system failures. Kennedy and Ghareeb (2017) [17] presented a literature review of different sucker rod artificial lift systems utilized in oil wells. The authors highlighted the limitations of conventional beam pumping equipment, including mechanical wear, inefficient pumping in gassy wells, and depth limitations. They also addressed common beam pumping system issues such as rod buckling, tubing wear, rod parting, and fluid pounding. Finally, they suggested several ideas to reduce failure frequency and extend run life, including using longer stroke lengths and slower strokes per minute, lowering peak polished rod loads, decreasing buckling tendencies, decreasing side loads and drag loads and long strokes to provide high-volume wells and handle gas problems. Khadav et al. (2018) [18] discussed the challenges and solutions for extending the life of rod-driven progressive cavity pumps (PCP) in the Bhagyam oilfield in India. They reported that installing centralizers with rod strings and using boronised tubing in high-wear areas led to longer tubing run life and less frequent workovers. Mahmoud Atef et al. (2019) [19] investigated the performance of artificial lift systems (ALSs) in heavy oil wells located in the eastern desert of Egypt. The study's objectives were to investigate the causes of ALS failures and to make recommendations for improving their performance. Research has focused on two forms of ALS: beam pumps (SRs) and progressive cavity pumps (PCPs).

Kumar and Upadhyay (2022) [20] discussed frequent tubing and sucker rod failures in deviated and vertical sucker rod pump (SRP) wells in an Indian oilfield operated by the Oil and Natural Gas Corporation (ONGC). These failures caused significant workover expenses and production losses. A total of 73.6% of all workover projects executed in 2018-19 were caused by SRP failure, with 44.9% due to rod and tubing failure. This study investigated the causes of these failures and proposed a detailed method for reducing and controlling tubing and rod failures. In this project, I propose an advanced solution to address the challenges associated with hole deviation, reducing the risk of failure due to tubing wear, rod parting, or other interventions caused by aggressive wellbore geometry. This approach will increase the average time between failures and change Egypt's oil and gas industry's mindset toward not considering the RRP or PCP as an ALS in highly deviated wells.

Data analysis

Well BED-3-26 Reservoir and fluid properties

A perforation interval between 10036 and 10085 ft was produced in BED-3-26, for which the total measured depth (MD) was approximately 10105 ft. The well yields approximately 250 barrels per day at a reservoir pressure of 3000 psi. The produced oil has an API gravity of 30 degrees, with a water cut of approximately 10%. The salinity of the water is 150,000 PPM, and the CO₂ content is approximately 1%. Table 1 provides a detailed overview of the reservoir fluid properties.

Table 1—Well BED-3-26 Reservoir and fluid properties

Characteristic	Value
Mid perforations measured depth	10060 ft
Wellhead pressure	120 PSI
Casing pressure	120 PSI
Water Cut	10%
GOR	500 SCF/STB
API Gravity	30 °API
H ₂ S content	3 PPM
CO ₂ content	1%
Water Salinity	150000 PPM
Bottom Hole Temp	233 °F
Static Level or pressure	3000 PSI
Productivity index	0.3 bpd/psi
Pump intake pressure	388 PSI
Sand content	2%

Well BED-3-26 Wellbore Geometry

Well BED-3-26 starts with a vertical section, followed by the kickoff point (KOP) at a measured depth (MD) of 4544 feet. The subsequent angle build section reaches the well's maximum angle of approximately 20.27°. Figure 2 shows the wellbore geometry in a vertical section of a typical well, while Figure 3 shows the MD versus hole angle. Figure 4 shows the MD and DLS, with a typical pay zone ranging from 0.5 to 8.44°/100 ft. The well was completed with a 13 3/8-inch surface casing, a 9 5/8-inch intermediate casing,

and a 7-inch production casing with a 400-foot sand rathole. Table 2 provides a detailed representation of the wellbore geometry.

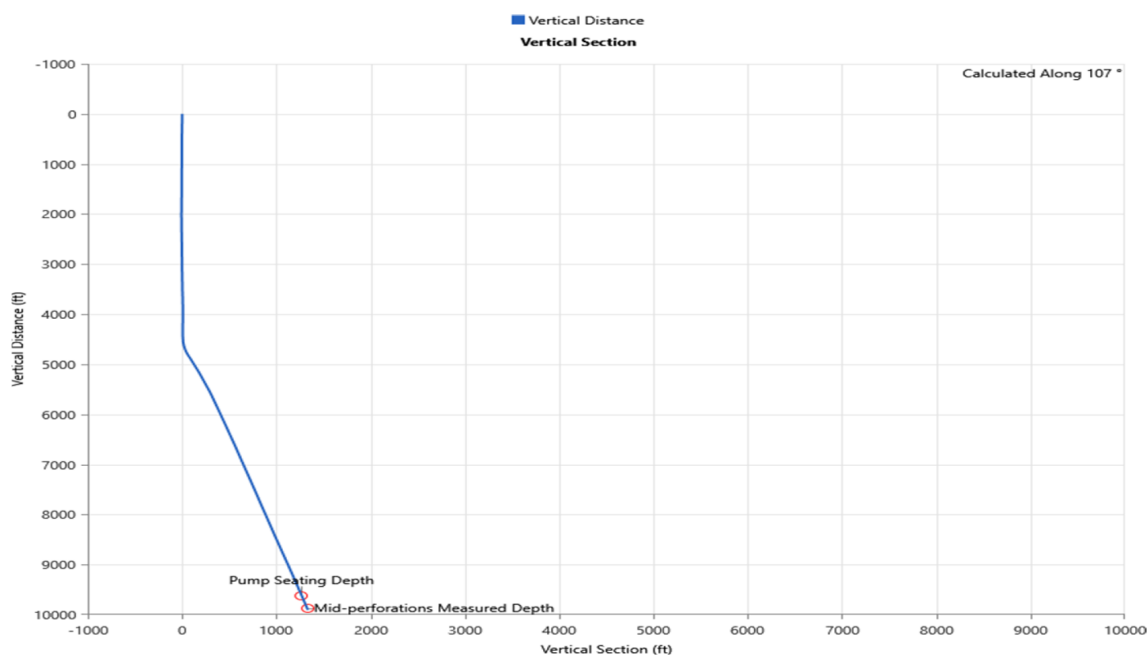


Figure 2—Well BED-3-26 MD vs. vertical section

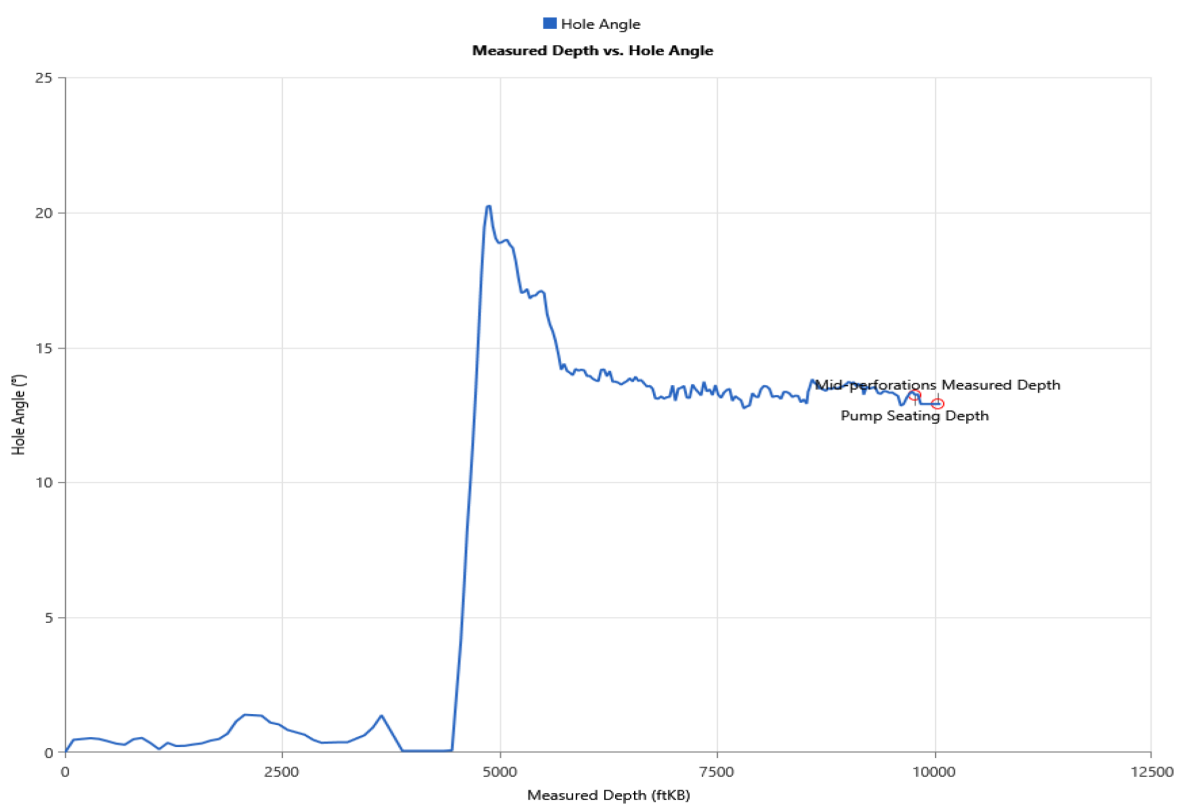


Figure 3—Well BED-3-26 MD vs hole angle

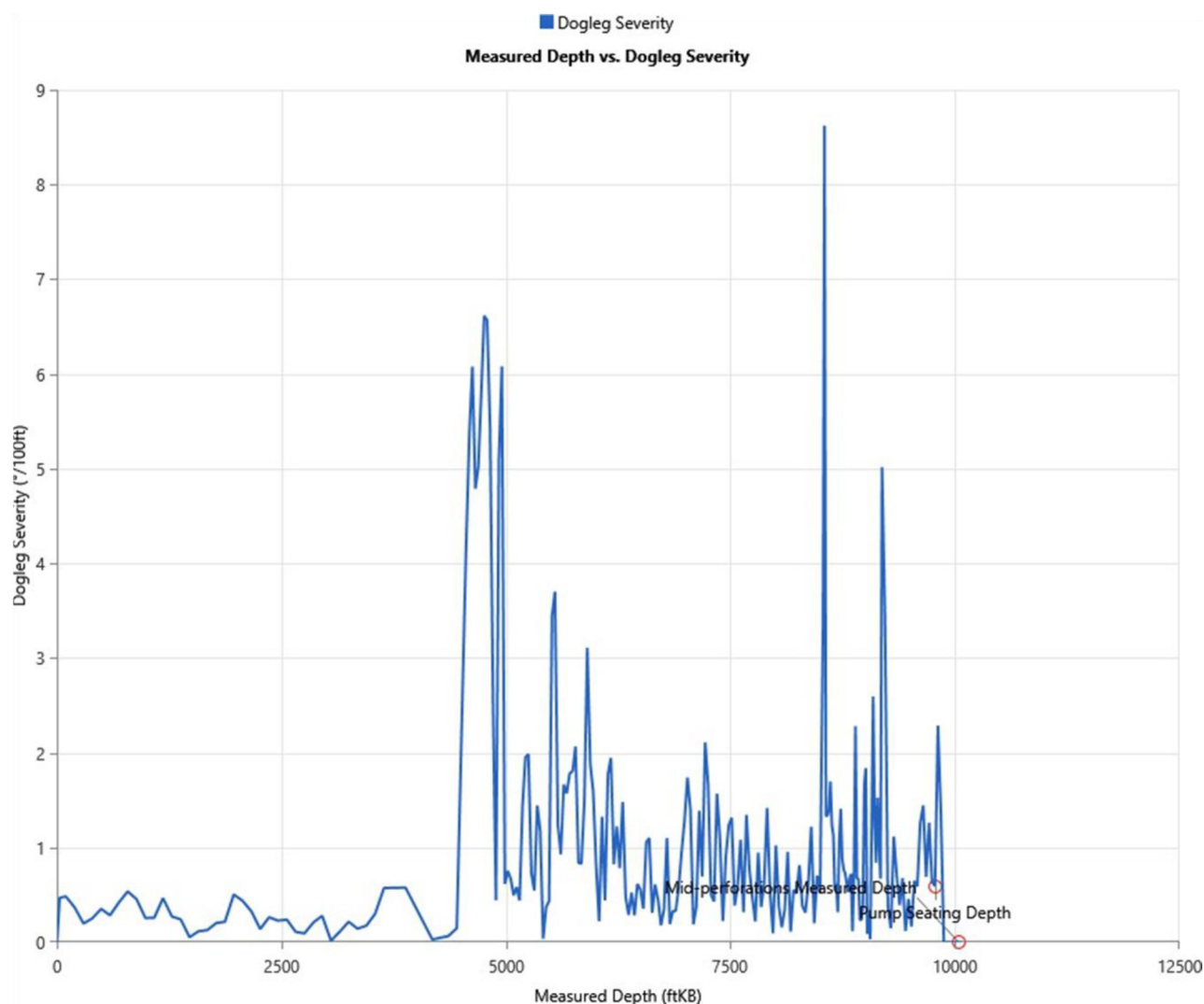


Figure 4—Well BED-3-26 MD vs DLS

Table 2—Well BED-3-26 wellbore geometry

Characteristic	Value
Max hole angle	20.27 °
Total measured depth	10105 ft
Max DLS	8.62 °/100 ft
Production casing	7 in
Tubing size	3-1/2 in

Well BED-3-26: Failure analysis

Any downhole failure in ESP applications necessitates a workover rig, which is both costly and operationally risky. Workover assessments have identified two primary risk categories, one of which is power cable damage during pulling or running, especially in deviated wells. In such wells, the wellbore trajectory causes the ESP power cable to rest against the low side of the casing due to gravitational forces. This results in sustained mechanical contact between the cable and casing wall during pulling operations, increasing the likelihood of abrasion and insulation damage, particularly in sections that are not properly clamped

or protected. Additionally, there is a risk of formation damage during well killing, especially in depleted reservoirs, where fluid losses can be significant. ESP operations in these environments are operated by high power consumption and frequent workovers, making them economically challenging. Given these constraints, the operator sought a more reliable and cost-effective artificial lift solution for this challenging application.

Potential Risks of Using Rod-Driven Systems in Highly Deviation Wells

One of the primary challenges of employing rod-driven systems in aggressive wellbore configurations is the frequent rod and tubing failures caused by high well deviation and DLS. These conditions increase the likelihood of rod-tubing contact, which, combined with the reciprocating motion of the rod string, can lead to rapid or severe wear and fatigue.

The wear rate and rod fatigue life are influenced by factors such as the DLS, final wellbore angle, pumping speed, and corrosion. Increased wear results in higher operational expenditure due to the need for frequent replacement of tubing and rods, as well as system downtime, which delays oil production and revenue. Additionally, complex workover procedures and fishing operations are often required to restore well production following tubing and rod failures.

Contact load

The contact load is the force applied to the rod string when it contacts the tubing. The contact load is determined by a variety of factors, including the type of rod string, well geometry, rod tension, well deviation, and the distribution of contact between the rod string and tubing. The couplings provide the majority of the contact between the tube and the sucker-rod string. Contact loads are caused by the combined forces of gravity and curvature, as illustrated in Figures 5 and 6 [22].

$$\begin{aligned}
 F_{CONTACT} &= F_{gravity} + F_{curvature} \\
 F_{gravity} &= \text{Gravity Induced Contact [N or lb]} \\
 F_{curvature} &= \text{Curvature Induced Contact Load [N or lb]}
 \end{aligned}
 \tag{Eq. 1}$$

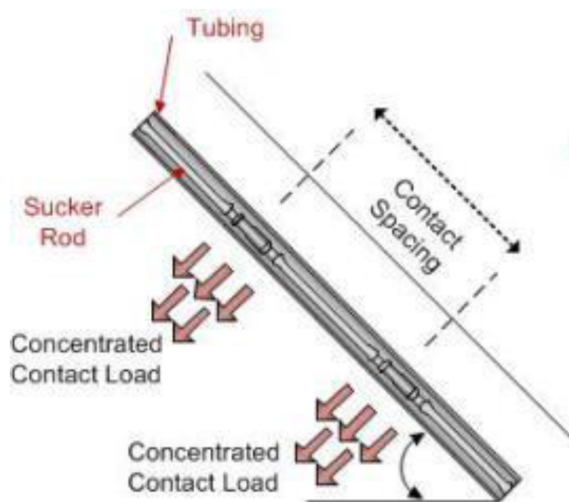


Figure 5—Gravity-induced contact load [22].

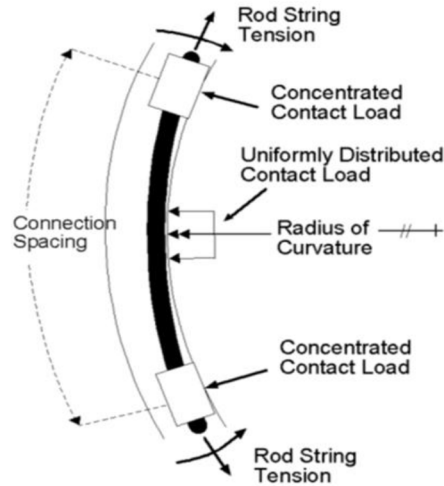


Figure 6—Curvature-induced contact load [22]

Gravity-induced contact load

$$F_{gravity} = C * W_r * L_{CS} * \sin\theta \quad \text{Eq. 2}$$

C = Constant [SI: 9.81 Imperial: 1]

W_r = Rod string unit weight [kg/m or lb/ft]

L_{CS} = Coupling spacing length [m or ft]

θ = Holte angle [°]

Curvature-induced contact load

$$F_{curvature} = C * L_{rod\ string} * L_{CS} * C_{wellbore} \quad \text{Eq. 3}$$

C = Constant [SI: 5.81×10^{-4} Imperial: 1.736×10^{-4}]

$L_{rod\ string}$ = Rod string tension [kg/m or lb/ft]

L_{CS} = Coupling spacing length [m or ft]

$C_{wellbore}$ = Wellbore curvature [°/(30m or °/100ft)]

What is the optimum solution for overcoming these challenges?

The combination of RRP systems with continuous rods is the key for reducing the risk of failure due to tubing wear, rod sectioning, or any other intervention caused by aggressive wellbore geometry. In addition, this combination will challenge the notion of not considering RRP or PCP systems as artificial lift systems in highly deviated wells. Compared to conventional sucker rods, continuous rods offer excellent replacement. The continuous rod only needs two rod-string coupling connections—one at either end of its length—compared to conventional sucker rods, which are coupled every 25 or 30 feet (7.6 or 9.1 meters). This unique uniformly distributed contact load product is ideal for highly deviated or horizontal wells because the rod stresses are distributed throughout the length of the rod string rather than concentrated on conventional sucker rod couplings, which must be coupled together on each conventional sucker rod. Figure 7 compares the continuous rod with the conventional sucker rod [22].

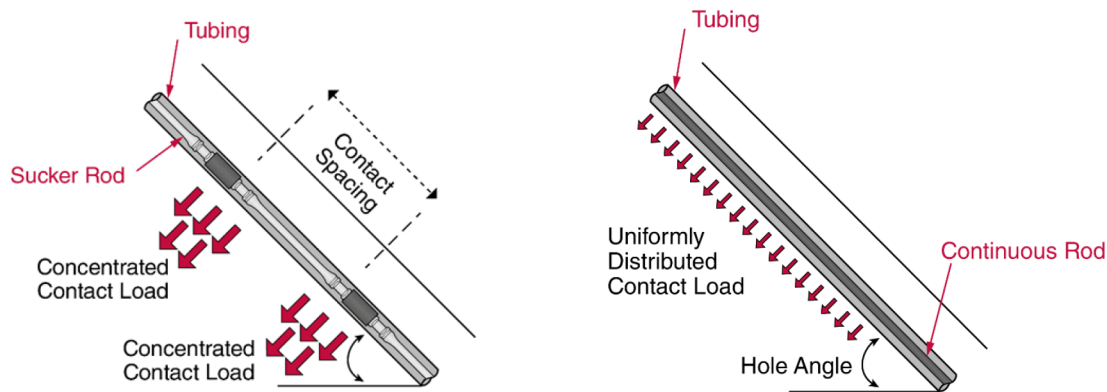


Figure 7—Comparison between conventional and continuous rod [22]

Continuous rod characteristics

- The uniform body design distributes contact loads equally along the whole rod, decreasing the severity of tubing and rod wear, as shown in Figure 8. [22].
- Regardless of the well depth, a continuous rod requires only two couplings, so there is no requirement for tightening or makeup torque. This eliminates weak spots that could lead to failure, as shown in Figure 9. [22].
- A continuous rod creates a larger tubing space, which reduces pressure losses by 50% to 75% compared to conventional sucker rods. This also increases production because the coupling piston effect is eliminated, especially when using the RRP system. Additionally, because there is no coupling, the flow switches from turbulent to laminar conditions by 25% to 50%, as shown in Figure 10 [22].
- A continuous rod is lighter than a conventional sucker rod string because no sucker rod couplings will be needed. This leads to less loading on the pumping unit and hence less power consumption [22].
- The run-in-hole time decreased due to the ease of operation and time savings when making rod connections. A continuous rod allows for the completion of two to three wells per day, resulting in a significant reduction in production delays [22].
- Because the coupling is stiffer than the rod body, the bending stresses are greater near the rod-coupling connections. However, a continuous rod reduces the rod stress because the rod curvature is equal to the wellbore curvature, as shown in Figure 11 [22].

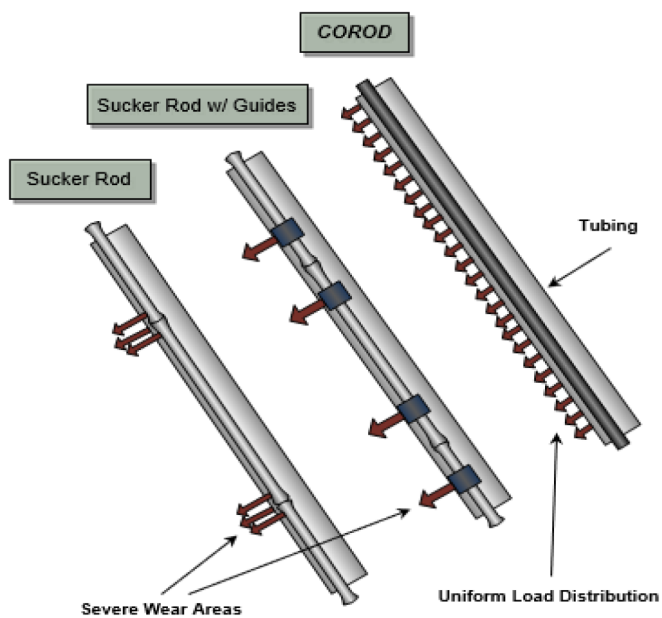


Figure 8—Contact load distributions for different types of sucker rod [22]

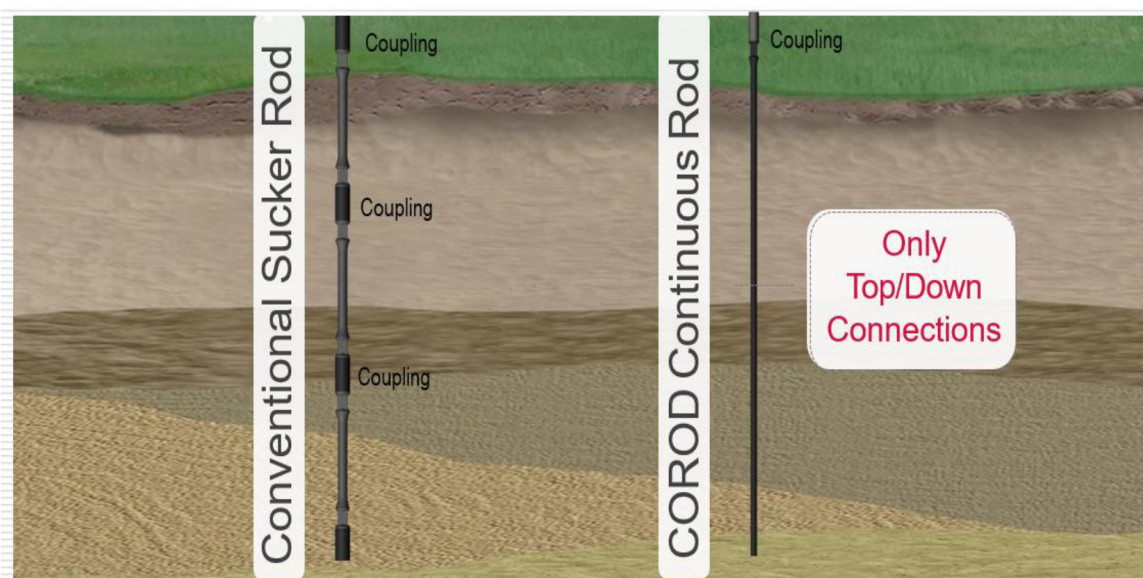


Figure 9—Thread connection comparison between conventional and continuous rod [22]



Figure 10—Comparison of the annular space between conventional and continuous rod [22]

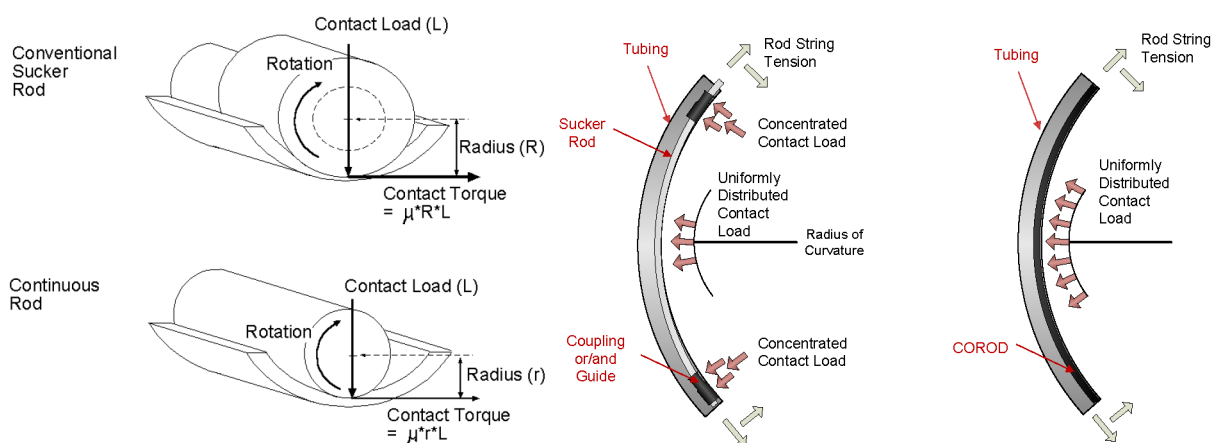


Figure 11—Continuous rod reduce rod stress and fatigue [22]

Well BED-3-26: Simulation study

A simulation study was conducted using software (CFER) to compare the performance of two sucker rod types (conventional sucker rods and continuous sucker rods), demonstrating the advantages of using continuous rods over conventional rods and how a continuous rod leads to reduced flow loss, reduced power consumption, and reduced contact between the sucker rod and tubing, resulting in decreased tubing wear vs. time [21].

Contact load prediction using a conventional sucker rod

CFER software uses the wellbore geometry input to estimate the contact loads by measuring the location of the rod string within the tubing along the whole length of the wellbore. The couplings are the points at which the maximum contact occurs between the sucker rod string and the tubing. Figure 12 shows the wellbore geometry and contact load distribution for BED-3-26. The maximum contact load is predicted to be 900 lb.

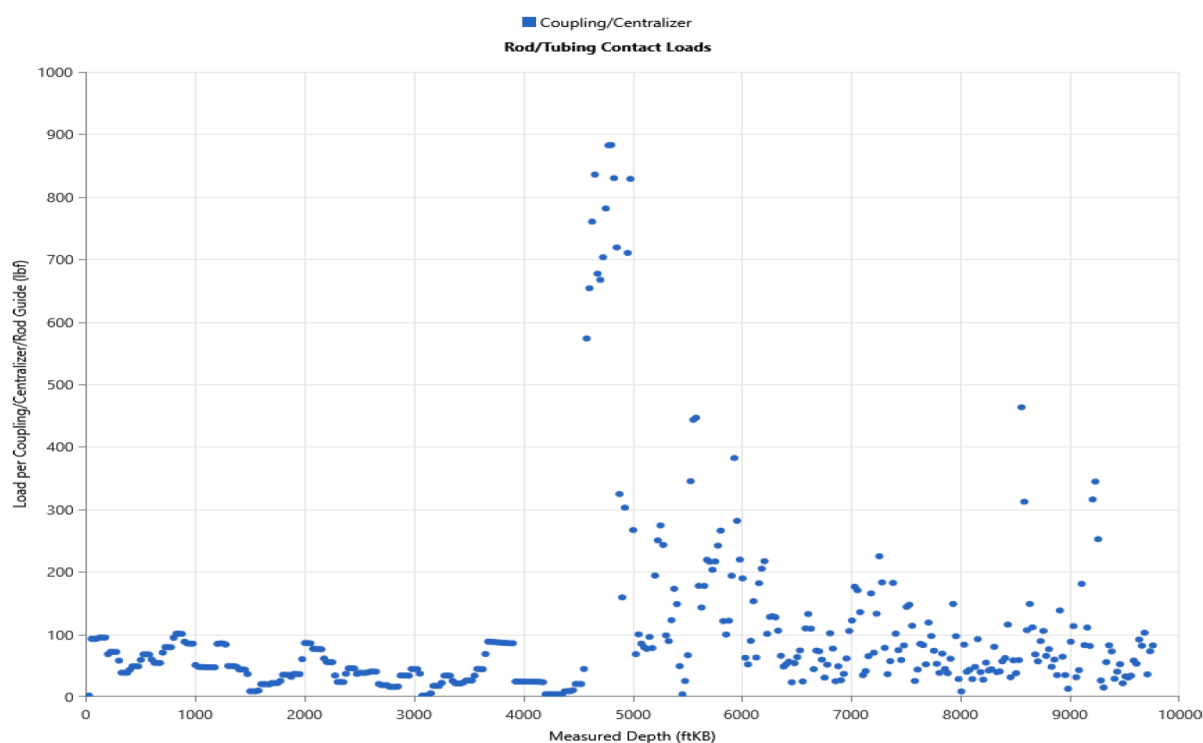


Figure 12—Well BED-3-26 Contact load estimation using conventional sucker rod

Well BED-3-26 Tubing wear prediction using conventional sucker rod

Friction between the tubing and rod caused by a high contact load, which increases on wellbore locations with high DLS values between the rod string and tubing, results in severe leakage and damage to both the tubing and sucker rod. Figure 13 shows the tubing wear rate vs. time using a conventional sucker rod. After 32 days, the tubing string is predicted to be completely worn.

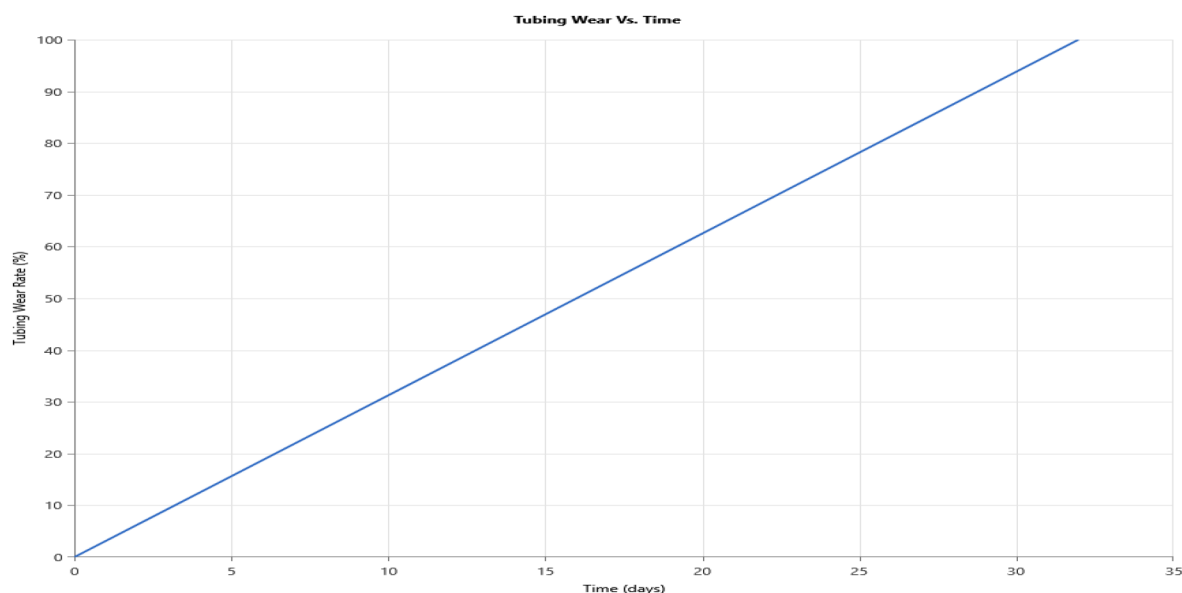


Figure 13—Well BED-3-26 Tubing wear vs. time with a conventional sucker rod

Well BED-3-26 Flow loss and power consumption predictions using a conventional sucker rod

Due to the presence of couplings, each 25 or 30 feet led to a decrease in tubing annular space, which resulted in an increase in flow losses, an increase in power requirements, and a decrease in the production rate.

According to the CFER software, the startup power input is approximately 50.59 hp, and the flow loss is approximately 71.74 psi when conventional sucker rods are utilized, as shown in Figure 14.

Analysis Results - Summary

Input Parameters		Output Parameters	
Fluid Properties		Basic Parameters	
Fluid Type	Single Phase	Pump Intake Pressure	388.89 psi
BS&W Content	12.00 %	Pump Discharge Pressure	4011.45 psi
Tubing Fluid Density	57.06 lbs/ft ³	Differential Pressure	3622.56 psi
Casing Fluid Density	57.06 lbs/ft ³	Net Hydrostatic Head	3400.81 psi
Gas Specific Gravity	0.70	Flow Losses	71.74 psi
Viscosity	100.00 cp	Pump Pressure Loading	85.24 % rated
Operating Conditions		Pump Torque	558.47 ft-lbs
Produced Fluid Flow Rate (surface, specified)	250.00 bbls/D	Pump Axial Load	11.57 kips
Produced Fluid Flow Rate (surface, specified)	250.00 bbls/D	Maximum Rod Torque	966.99 ft-lbs
Pump Speed	211.42 RPM	Maximum Rod Torque Load	70.03 % Limit
Surface Motor Operation Frequency	43.28 Hz	Maximum Axial Rod Load	39.89 kips
Pump Volumetric Efficiency	80.00 %	Maximum Effective Rod Stress	87.88 %
Fluid Level	8776.90 ftKB	Rod Contact Loads	
Submergence (MD) (Specified)	1000.00 ftKB	Maximum Load Connection	723.32 lbf
Submergence (TVD)	972.75 ftKB	Surface Parameters	
Bottomhole Pressure	490.35 psi	System Input Power	50.59 hp
Tubing Head Pressure	150.00 psi	Prime Mover Output Power	45.53 hp

Figure 14—Well BED-3-26 Tubing wear estimation using a conventional sucker rod

Well BED-3-26 Contact load prediction using a continuous rod

As shown in Figure 15, replacing the conventional rod with a continuous rod reduced the side load to 33 lb. from 900 lb. The contact load of a continuous rod is 25-30 times lower than that of a conventional rod coupling. This reduces the potential for failure due to tubing leak, rod parting, or any other intervention caused by aggressive wellbore geometry.

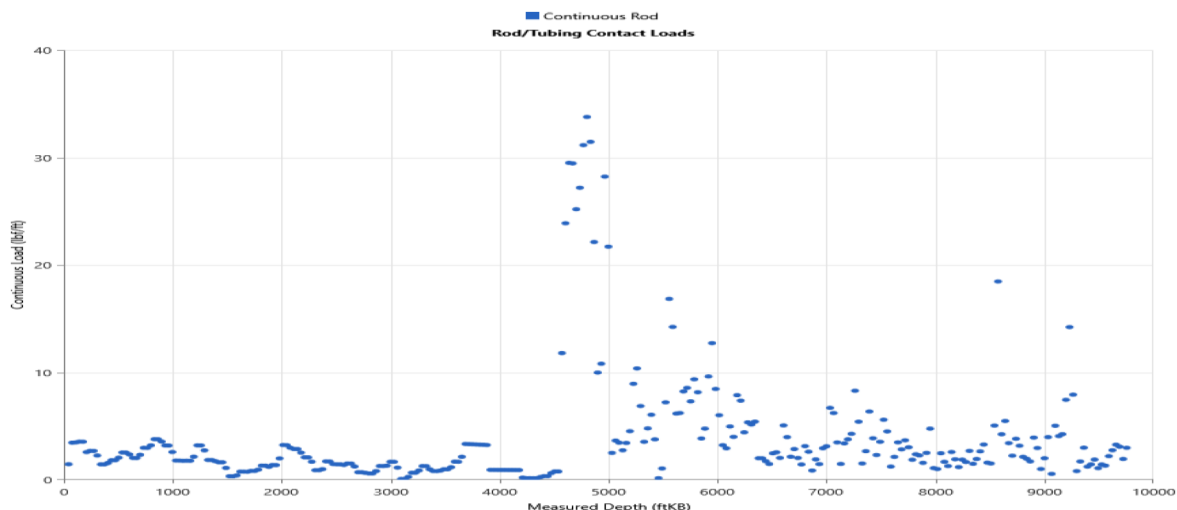


Figure 15—Well BED-3-26 Contact load estimation using continuous rod

Well BED-3-26 Tubing wear prediction using continuous rod

Using a continuous sucker rod rather than a conventional rod is the best approach. The uniform body and even contact distribution of this type of sucker rod between the tubing and the rod string have been demonstrated in the field to significantly reduce wear-related problems. Figure 16 shows a tubing wear vs. time example when employing a continuous rod. The tubing wear rate vs. time increased to 500 days, compared to 32 days for conventional sucker rods. This reduces the potential for failure due to tubing wear, rod failure, or other interventions caused by aggressive wellbore geometry.



Figure 16—Well BED-3-26 Tubing wear estimation using continuous rod

Well BED-3-26 Flow loss and power consumption predictions using a continuous rod

Continuous rod minimize flow losses by 40-75% compared to conventional sucker rods due to the greater tubing annular space. The elimination of the coupling piston effect, particularly when using the RRP system, increases production and decreases power usage. The CFER software predicts that using a continuous rod will result in flow losses of approximately 42.66 psi and power requirements of 39.43 hp, as illustrated in Figure 17.

Analysis Results - Summary

Input Parameters		Output Parameters	
Fluid Properties		Basic Parameters	
Fluid Type	Single Phase	Pump Intake Pressure	388.89 psi
BS&W Content	12.00 %	Pump Discharge Pressure	3982.36 psi
Tubing Fluid Density	57.06 lbs/ft ³	Differential Pressure	3593.47 psi
Casing Fluid Density	57.06 lbs/ft ³	Net Hydrostatic Head	3400.81 psi
Gas Specific Gravity	0.70	Flow Losses	42.66 psi
Viscosity	100.00 cp	Pump Pressure Loading	84.55 % rated
Operating Conditions		Pump Torque	554.63 ft-lbs
Produced Fluid Flow Rate (surface, specified)	250.00 bbls/D	Pump Axial Load	11.48 kips
Produced Fluid Flow Rate (surface, specified)	250.00 bbls/D	Maximum Rod Torque	753.80 ft-lbs
Pump Speed	211.42 RPM	Maximum Rod Torque Load	63.06 % Limit
Surface Motor Operation Frequency	43.28 Hz	Maximum Axial Rod Load	36.88 kips
Pump Volumetric Efficiency	80.00 %	Maximum Effective Rod Stress	84.16 %
Fluid Level	8776.90 ftKB	Rod Contact Loads	
Submergence (MD) (Specified)	1000.00 ftKB	Maximum Load Rod	27.08 lbf/ft
Submergence (TVD)	972.75 ftKB	Surface Parameters	
Bottomhole Pressure	490.35 psi	System Input Power	39.43 hp
Tubing Head Pressure	150.00 psi	Prime Mover Output Power	35.49 hp

Figure 17—Well BED-3-26 Tubing wear estimation using continuous rod

Well BED-3-26 simulation results

Two scenarios (continuous rods without couplings and standard sucker rods with couplings) are compared to estimate rod contact with tubing and tubing wear over time. Figure 18 and Table 3 depict the simulation results obtained using the CFER prediction software. The CFER software predicts that utilizing continuous rods delays the tubing total wear to 500 days, compared to 32 days for conventional sucker rods, while reducing the side load to 33 lb. from 900 lb., reducing flow losses from 71.74 psi to 42.66 psi and reducing the power usage from 50.59 hp to 39.43 hp.

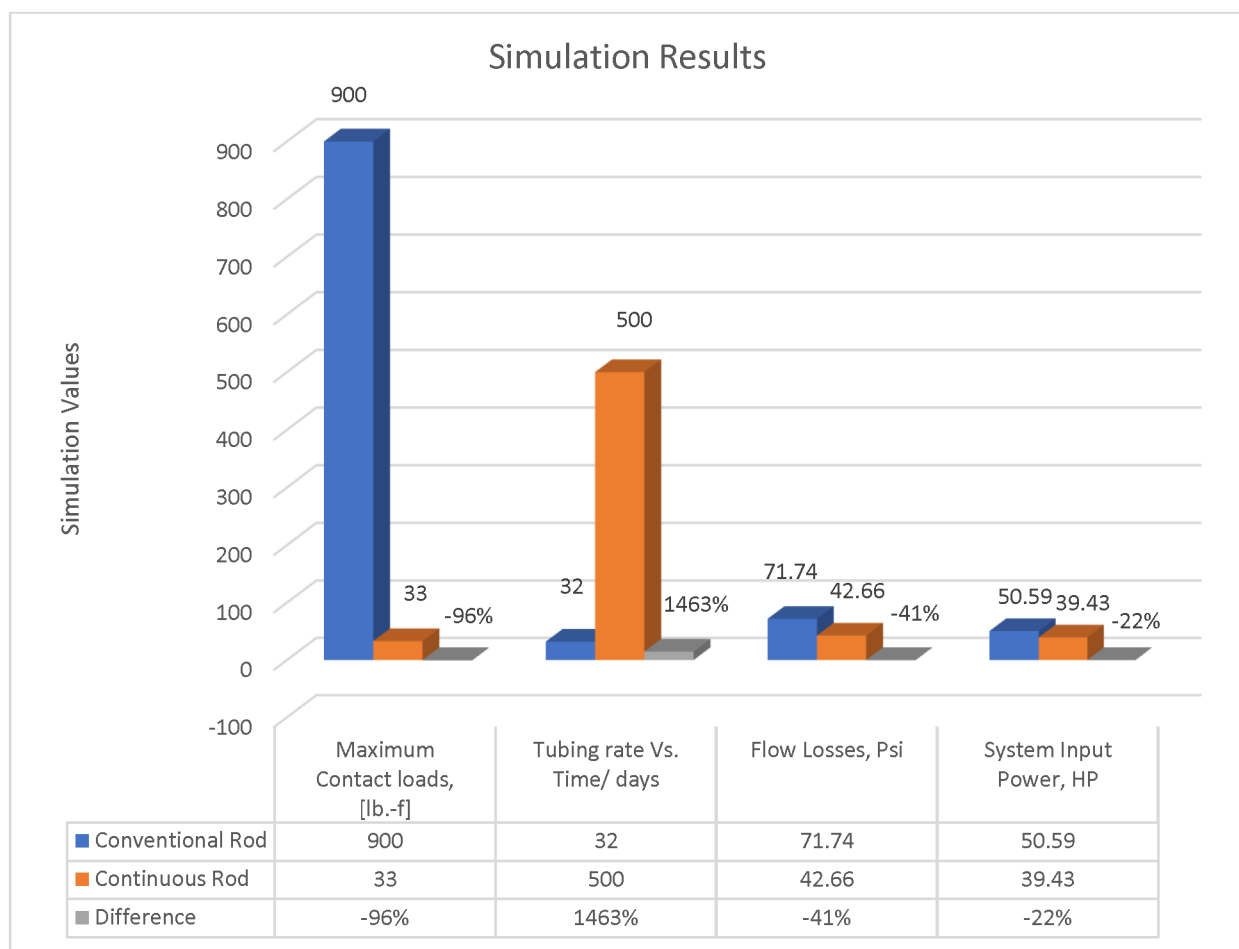


Figure 18—Well BED-3-26 simulation results

Well BED-3-26 Field results

The utilization of RRP system combined with continuous rods has been in continuous operation for almost 460 days without any intervention, whereas the maximum run life of 90 days was achieved when ESP was placed in the well. Figure 19 demonstrates the actual runtime between the two ALS systems. The graph demonstrates that RRP combined with a continuous rod increased the well's run life by 411% over its previous run life when a costly ESP was used.

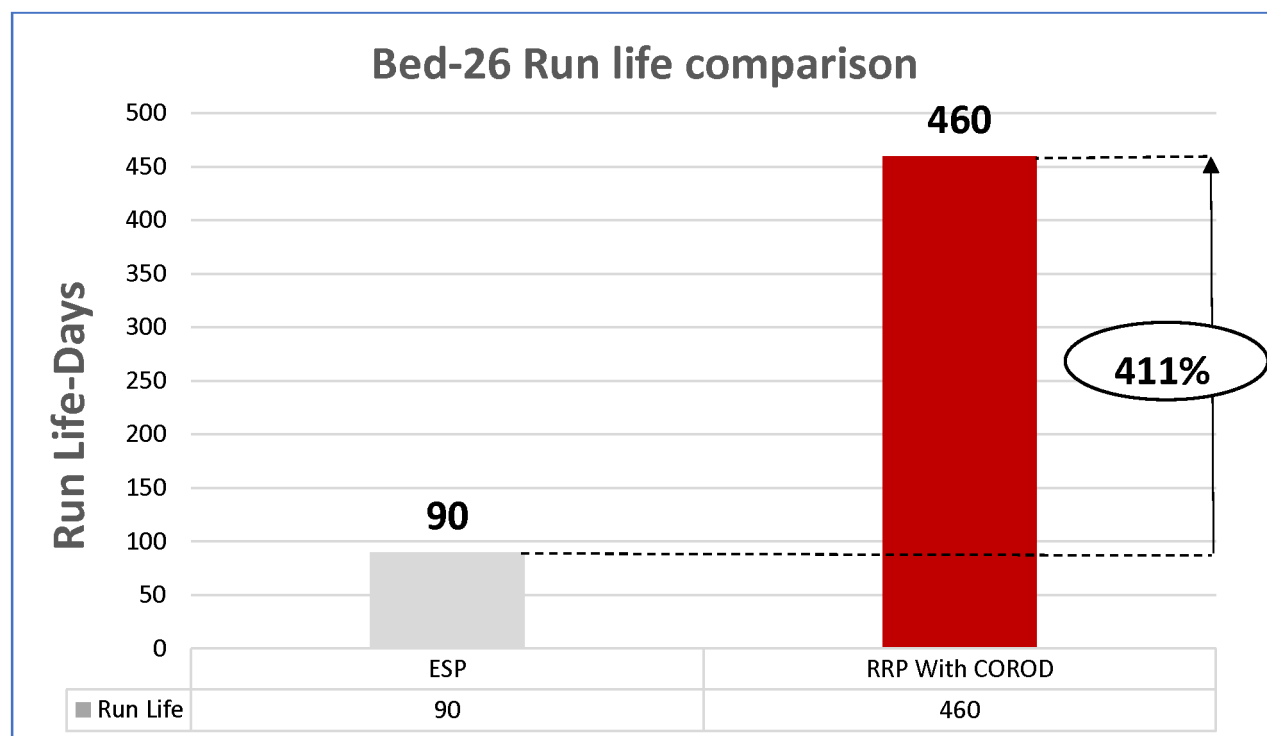


Figure 19—Well BED-3-26 Run life comparison

Well BED-3-26 Continuous Rod Operational benefits

- Expected tubing run life has been increased from 32 to 500 days by combining the RRP system with a continuous rod, compared to 32 days for conventional sucker rods. This work can guide the evaluation of the expected service life of continuous rod installation against that of conventional sucker rod installation.
- The operator benefited from continuous rod strings, which decreased well servicing and rod/tubing replacement expenses, avoided production losses, and increased the average time between failures from 90 to 460 days. Over 411% is the recorded run-life enhancement over the well's previous run-life. The well had previously failed four times in an average year.
- Workover frequency has decreased by more than 100%.
- Combining the RRP system with a continuous rod decreased the operator's initial CAPEX investment by 39%.
- The RRP system with a continuous rod reduced horsepower usage by approximately 70% compared to that of the previous ESP placed in the well. The horsepower needed was reduced from 170 HP to 100 HP.
- Employing a continuous rod reduced flow losses by 41% due to increased annular tubing space.
- Continuous rod strings minimize the environmental impact by reducing the power consumption and rod replacement frequency.
- A continuous rod increases well productivity by eliminating the coupling piston effect, especially in RRP systems.
- Eliminating sucker rod coupling reduces the rod string weight by 8%, reducing the loading on surface equipment and requiring smaller, lower-cost equipment.
- Eliminating sucker rod connections reduces pin-related sucker rod failures and improves quality through reduced human handling.

- The main factor in lowering system CAPEX and OPEX costs in these challenging applications is the implementation of a continuous rod.

Well BED-3-26 Commercial results

As shown in Table 4, The commercial analysis showed that the RRP system combined with continuous rods provides considerable financial benefit to operators in wells with high DLSs that were previously considered to be low- and medium-flow ESP devices. This shift from costly, high-power ESP to RRP combined with a continuous rod resulted in overall savings of MM 3,8\$

Table 4—Well BED-3-26 Commercial result.

Well Depth, M		2980
Problem Well Economics	Assumptions of problem well	
	ESP system CAPEX	\$250,000
	Number of interventions per well per year	4
	Number of days waiting for rig per intervention	15
	Number of rig working days per intervention	3
	Average production bbl./day	250
	Deferred production bbl. per year	18000
	Price per barrel	\$75
	Cost of deferment	\$1,350,000
	Workover cost per day	\$5,000
	Total workover cost per year	\$60,000
	Cost of equipment per intervention (60 % of ESP Cost)	\$150,000
	Total replacement equipment per intervention	\$600,000
	Diesel consumption for ESP Liter/day.150 HP	260
	Diesel price /Liter	\$1
	Power consumption for ESP system per year-150 HP	\$76,180
	Total cost per year	\$2,336,180
	Total revenue per year	\$6,843,750.00
	Total revenue per year excluding total cost	\$4,507,570.00
Cost of New Technology	Cost of alternative Technology	
	RRP with Continuous rod system CAPEX	\$180,000
	Average production bbl./day	250
	Price per barrel	\$75
	Diesel consumption for RRP with Continuous rod -Liter/day.	140
	Diesel price /Liter	\$1
	Power consumption for RRP with Continuous rod per year -100 HP	\$51,100
	Total cost per year	\$231,100
	Total revenue per year	\$8,625,000.00
	Total revenue per year excluding total cost	\$8,393,900.00
Total Savings for Well BED-3-26 per year		\$3,886,330.00

Conclusions and Recommendations

- The implementation of continuous rods in Well BED-3-26 has challenged the conventional wisdom that RRP systems are unsuitable for highly deviated wells.
- Adopting RRP or PCP systems in combination with continuous rods offers a cost-effective alternative to expensive, high-power ESP systems, thereby reducing lifting costs and accelerating return on investment in Egypt's severely deviated wells.
- This study demonstrated that continuous rods are among the most effective and optimized solutions for reciprocating rod lift (RRL) in Egypt. The deployment of continuous rods in this challenging well is projected to save nearly \$4 million annually.
- Continuous rods combined with an RRP system can increase the average time between failures by 411%, reduce flow losses by 41%, lower the initial CAPEX by 39%, and decrease the power consumption by 70%.

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Abbreviations

ESP	Electrical Submersible Pumping
DLS	Dog Leg Severity
Psi	Pound per square inch
MD	Measured depth
PPM	Parts per million
%	Percent
M	Meter
°F	Fahrenheit degree
SCF/STB	Standard cubic feet per stock tank barrel
API	American Petroleum Institute
bpd/psi	barrel per day/Pound per square inch
KOP	Kickoff point
Vs	Versus
°	Degree
Ft	Feet
in	inch
lb	pound
ALS	Artificial Lift Systems
BAPETCO	Badr El Din Petroleum Company
CAPEX	Capital Expenses
EGPC	Egyptian Petroleum Corporation
BFPD	Barrel fluid per day
PCP	Progressing Cavity Pumping
OPEX	Operating Expenditure

RRP	Reciprocating Rod Pumping
WC	Water cut.
θ	Hole angle
bbl/STB	Barrel per stock tank barrel
C	Constant
Fcurvature	Curvature-Induced Contact Load
Fgravity	Gravity-Induced Contact Load
Lcs	Coupling spacing length.
Lbf	Pounds force
Lrod string	Rod string tension
MM	Million
Cwellbore	Wellbore curvature
ONGC	Oil and Natural Gas Corporation

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